



Naman Karwa

Roll No.:230123037

B.Tech - Mathematics and Computing
Indian Institute Of Technology, Guwahati

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.45 (Current)	2023-Present
Senior Secondary	CBSE Board	94.4%	2023
Secondary	CBSE Board	96.6%	2021

EXPERIENCE

• Research Internship

June 2025 - Present

Under Dr. Joy Das, Cornell University

- Initiated collaborative research exploring the impact of **SEBI's agri-derivative trading bans** on food inflation in India, focusing on the price dynamics of essential grains.
- Reviewing relationship between agricultural commodity futures and consumer price inflation; currently evaluating modeling approaches such as **GARCH, Granger causality, and synthetic control methods** to analyze these effects.

PROJECTS

• Trading Strategy for BTC/USDT and ETH/USDT Market

Oct. 2024 - Dec. 2024

Inter IIT Tech Meet 13.0, IIT Bombay

- Developed comprehensive crypto trading strategies for BTC/USDT and ETH/USDT markets, leveraging advanced **Reinforcement Learning** models for BTC and in-depth **correlation analysis** for ETH price movements.
- Utilized novel analytical approaches like **CUSUM analysis** and **Q-learning** to optimize decision-making processes and enhance strategy robustness under volatile conditions.
- Achieved a net profit of **224% for BTC/USDT (beating the benchmark by 42.68%)** and **7684% for ETH/USDT (beating the benchmark by 355.48%)** with respective win rates of **68% and 48%**.

• Pairs Trading: Statistical Arbitrage using Cointegration

Nov. 2024

Finance and Economics Club, IIT Guwahati

- Developed a market-neutral equity pairs trading strategy leveraging **high-correlation filtering** and **Engle-Granger cointegration tests** to identify statistically stable equity pairs from Nifty stocks.
- Executed historical backtesting over 5-year daily data, achieving **113.93% cumulative return, 11.89% CAGR, Sharpe Ratio 1.23, Sortino Ratio 1.94, and Max Drawdown -13.71%** across 430+ trades.
- Applied dynamic hedge ratio modeling and **z-score-based mean reversion signals** to trigger long-short trades on cointegrated pairs with significant price spread deviations.

• Black-Litterman Model for Optimized Portfolio Allocation

Sept. 2024

Finance and Economics Club, IIT Guwahati

- Constructed a robust **market-driven portfolio** by analyzing historical stock returns, computing excess returns over the risk-free rate, and deriving implied equilibrium returns using market-cap weights.
- Incorporated subjective investor views by constructing a **view matrix (P)** and **uncertainty matrix**, quantifying confidence levels in specific market predictions and asset relationships.
- Applied **Bayesian updating** to blend market equilibrium with investor views, optimizing portfolio weights using mean-variance framework for enhanced risk-adjusted returns.

TECHNICAL SKILLS

• **Programming:** Python, C/C++

• **Data Analysis:** Numpy, Pandas, Matplotlib, Scikit-learn, Yfinance, TA-lib

• **Database Management:** MySQL

KEY COURSES TAKEN

• **Mathematics:** Linear Algebra, Basic Calculus, Discrete Mathematics, Probability & Random Processes

• **Quantitative Finance:** Portfolio Optimization, Fundamental Analysis, Technical Analysis

• **Computer Science:** Data Structure, Digital Design, Database Management, Computer Organization & Architecture

ACHIEVEMENTS

• **Gold Medal, Inter IIT TechMeet 2024-25** in UnTrade Zelta Automations Problem Statement, held at IIT Bombay

• **AIR 1839, JEE Main 2023** among 1.2 million candidates appearing in the examination

• **AIR 1608, JEE Advanced 2023** among 0.2 million candidates appearing in the examination

POSITION OF RESPONSIBILITY

• Trading and Strategy Head, Finance and Economics Club, IIT Guwahati

April 2025 - Present

- Leading a team of 40 associates; designing and delivering curriculum on Technical Analysis, Backtesting, & Options trading with virtual and live market simulations to enhance practical understanding.

• Branch Representative, Mathematics and Computing'23

April 2024 - Present

- Efficiently managed and organized activities for a batch of 80+ students, effectively addressing their issues.
- Acted as the primary contact point for professors and provided peer tutoring support to fellow students.